

Lecture 17, May 04 2026.

11.11 Applications of Taylor Polynomials.

let $f(x)$ is a function then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Let

$T_n(x) :=$ the n th partial sum of this series:

this is the n th-degree Taylor polynomial of f at a .

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + \frac{f'(a)}{1!} (x-a)^1 + \frac{f''(a)}{2!} (x-a)^2 + \dots +$$

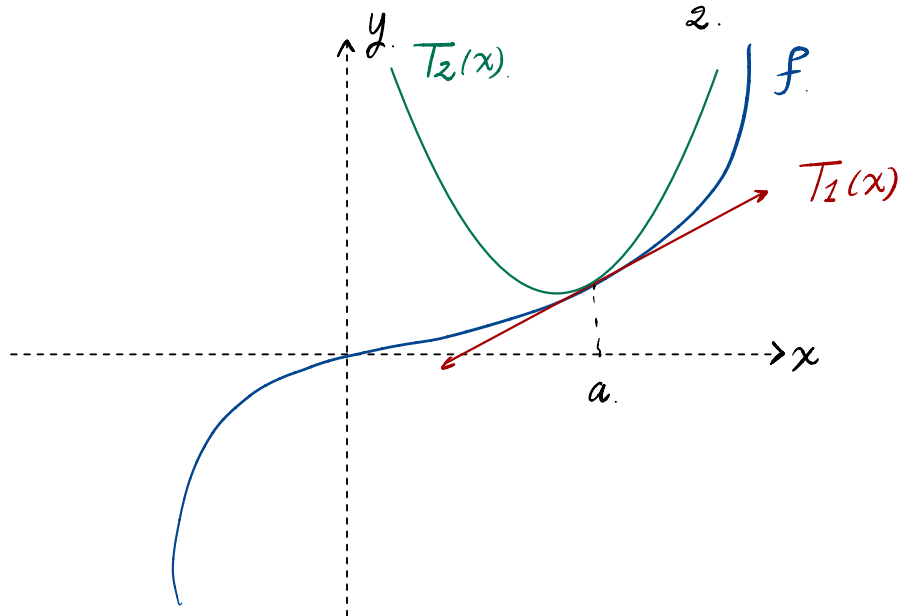
$$\frac{f^{(n)}(a)}{n!} (x-a)^n.$$

if $n \rightarrow \infty$ then $T_n(x) \rightarrow f(x)$; so, $T_n(x) \approx f(x)$.

if $n=1$, the first-degree Taylor polynomial

$$T_1(x) = f(a) + f'(a)(x-a)$$

$$T_2(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2} (x-a)^2$$



Rmk. $f(x) = T_n(x) + R_n(x)$.

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}_{T_n(x)} + \underbrace{\sum_{k=n+1}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k}_{R_n(x)}.$$

and

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$$

Example. a). Approximate the function $f(x) = \sqrt[3]{x}$ by Taylor polynomial of degree 2 at $a=8$.

b). How accurate in this approximation when $7 \leq x \leq 9$?

Sln. a). $f(x) = \sqrt[3]{x} \Rightarrow f(8) = 2.$

$$f'(x) = \frac{1}{3} x^{-2/3} \Rightarrow f'(8) = \frac{1}{12}$$

$$f''(x) = -\frac{2}{9} x^{-5/3} \Rightarrow f''(8) = -\frac{1}{144}$$

$$T_2(x) = f(a) + \frac{f'(a)}{1!} (x-a)^1 + \frac{f''(a)}{2!} (x-a)^2$$

$$= 2 + \frac{1}{12} (x-8) - \frac{1}{288} (x-8)^2$$

b). The Taylor series is not alternating when $x < 8$, so we can't use the Alternating Series estimation in this example.

if $n=2$, $a=8$.

$$|R_2(x)| \leq \frac{M}{3!} |x-8|^3$$

where $|f'''(x)| \leq M$.

$$f'''(x) = \frac{10}{27} \cdot \frac{1}{x^{8/3}} \quad \text{so, } x^{8/3} \geq 7^{8/3} \quad \text{so}$$

$$f'''(x) = \frac{10}{27} \cdot \frac{1}{x^{8/3}} \leq \frac{10}{27} \cdot \frac{1}{7^{8/3}} \quad \left\{ \begin{array}{l} f'''(7) = M, \text{ if } c \leq x \leq d \\ \text{then } M = f^{(n+1)}(c) \end{array} \right.$$

$$\approx 0.0021$$

Therefore,

$$|R_2(x)| \leq \frac{0.0021}{3!} |7-8|^3 = \frac{0.0021}{6} < 0.0004$$

Thus, if $7 \leq x \leq 9$, the approximation in part a). is accurate to within 0.0004.

Example. a). what is the maximum error possible in using the approximation

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

when $-0.3 \leq x \leq 0.3$? Use this approximation to find $\sin 12^\circ$ correct to six decimal places.

b). For what values of x is this approximation accurate within 0.00005.

Sln: a). We know that

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Since $|x| < 1$ So, we can use the A-S estimation.

$$\left| \frac{x^7}{7!} \right| = \frac{|x|^7}{5040}$$

So, if $-0.3 \leq x \leq 0.3$ So the error is smaller than

$$\frac{(0.3)^7}{5040} \approx 4.3 \times 10^{-8}$$

To find $\sin 12^\circ$ we first convert to radian measure:

$$\sin 12^\circ = \sin \left(\frac{12 \pi}{180} \right) = \sin \left(\frac{\pi}{15} \right).$$

$$\approx \frac{\pi}{15} - \left(\frac{\pi}{15} \right)^3 \cdot \frac{1}{3!} + \left(\frac{\pi}{15} \right)^5 \cdot \frac{1}{5!} \approx 0.20791169$$

b). The error will be smaller than 0.00005 if

$$\frac{|x|^7}{5040} < 0.00005.$$

Solving

$$|x| < \sqrt[7]{(0.00005)(5040)} \approx 0.821.$$

